

02_VEHICLE_SPEED_CALIBRATION

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EVIDENCE PROVENANCE: [PHYSICAL (DERIVED)]
Derived from 10-second optical tachometer staircase calibration runs under load on dual prototype chassis.

1. Definitive Terminology: 34.58 is K_cal

Vehicle A has 42 physical slots; Vehicle B has 43 physical slots. Under load, micro-slip and debounce yield an empirical divisor of 34.58 pulses/rev. It is NOT physical PPR; it is K_cal (Empirical Calibration Divisor).

2. Calibration Parameters Truth Table

Parameter	Vehicle A (TRUCK_01)	Vehicle B (TRUCK_02)	Provenance
Motor Driver	L298N Bipolar	TB6612FNG MOSFET	HARDWARE
Wheel Diameter (D)	0.060 m (60 mm)	0.060 m (60 mm)	PHYSICAL (0.060m)
Physical Slots (PPR)	42 slots	43 slots	PHYSICAL INSPECTION
Calibration Divisor K_cal	34.58 pulses/rev	34.58 pulses/rev	PHYSICAL (DERIVED)
Linear Resolution	5.451 mm/pulse	5.451 mm/pulse	DERIVED: C / K_cal
Safe PWM Ceiling	220 (out of 255)	240 (out of 255)	THERMAL LIMIT
Calibrated V_max	1.40 m/s	1.30 m/s	PHYSICAL STAIRCASE
Road Baseline Limit	0.80 m/s	0.80 m/s	CONFIGURED LIMIT